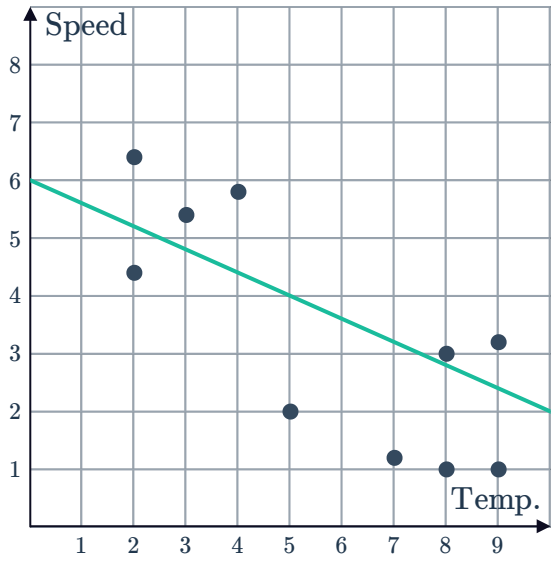
The least squares line of best fit

1 The average monthly temperature in $^\circ C$, and the average wind speed in knots, in a particular location was plotted over several months. The graph shows the points for each month's data and their line of best fit. Use the line of best fit to approximate the wind speed on a day when the temperature is $5^\circ C$.



2 A plane's altitude (A) is measured at several times (t) during its descent. The data and the line of best fit are shown below.

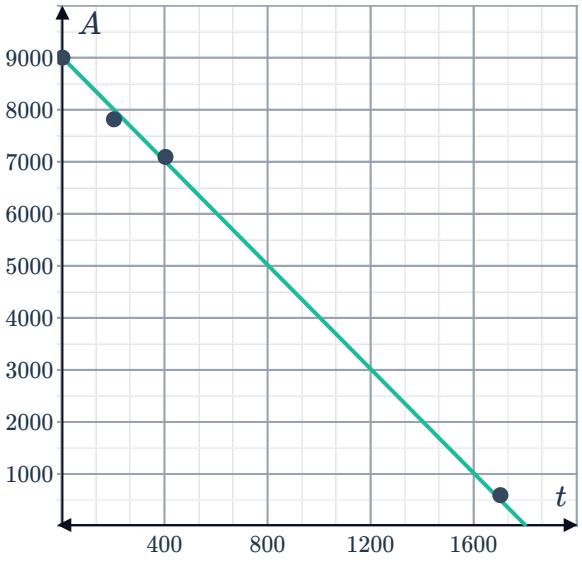
Time (t seconds)	0	200	400	1700
Altitude (A metres)	9000	7815	7092	593

a According to the graph, what is the altitude of the plane 100 seconds into the descent?

b According to the graph, what is the altitude of the plane 500 seconds into the descent?

c According to the graph, for how many seconds has the plane been descending when it is at an altitude of 7500 metres?

d According to the graph, how many seconds did the plane take to descend to the ground?



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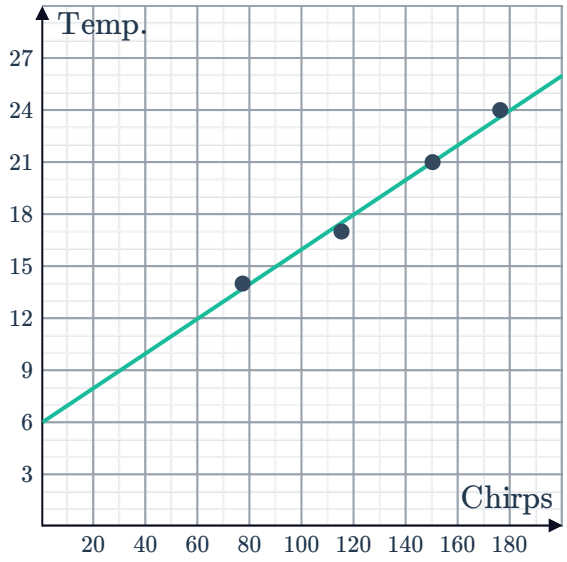
3 Chirping crickets can be an excellent indication on how hot or cool it is outside. Different species of crickets have different chirping rates but for a particular species the following data was recorded:

Number of chirps per minute	77	115	150	176
Temperature (°C)	14	17	21	24

a According to the graph, what is the temperature when the crickets make 140 chirps each minute?

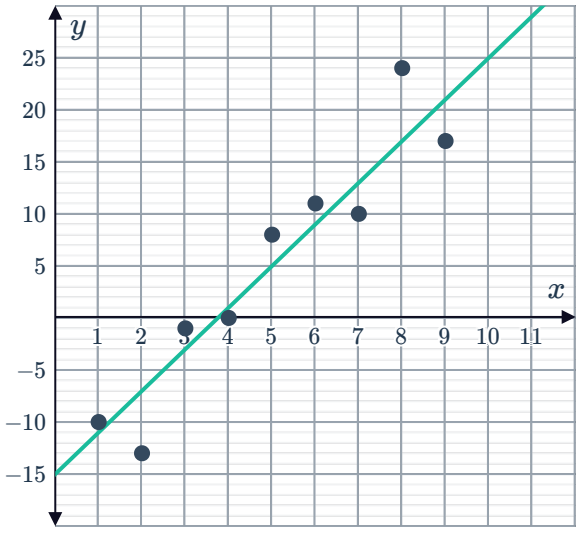
b According to the graph, how many chirps per minute will the crickets make if the temperature is 27?

c According to the graph, how many chirps are the crickets making each minute if the temperature is 19°C?



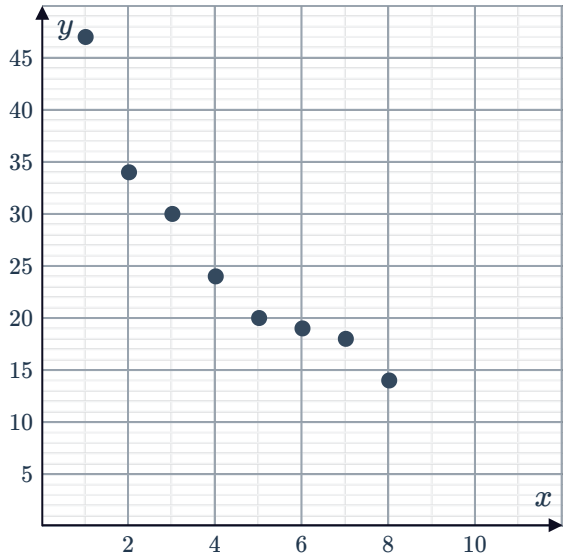
The equation of a least squares line of best fit

4 Find the equation for the line of best fit shown:



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5 Consider the following scatter plot:



a Is the relationship between the x and y variables positive or negative?

b Which of the following could be the equation for the line of best fit?

A $y = -4x - 4$ **B** $y = 44 + 4x$

C $y = -4x + 44$ **D** $y = 4x - 4$

6 The equation $d = 58 - 0.63h$ represents the line of best fit relating the air humidity, h, and the depth in metres, d, of snow in an area.

a Use the equation to determine the snow depth when the air humidity is 0.6.

b Find h, the level of air humidity you would expect to achieve a snow depth of 57.496 metres.

7 The equation for the line of best fit is given by $P = 161 - 2t$, where t is time.
Over time, is P increasing, decreasing or remaining constant?

8 The table shows the number of people who went to watch a movie x weeks after it was released:

Weeks (x)	1	2	3	4	5	6	7
Number of people (y)	17	17	13	13	9	9	5

a Plot the points from the table on a number plane.

b Graph the line of best fit whose equation is given by $y = -2x + 20$ on the same number plane.

c Use the equation of the line of best fit to find the number of people who went to watch the movie 10 weeks after it was released.

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9 The table shows data on the number of kilograms of litter collected each week in a national park x weeks after the park managers started an anti-littering campaign:

Weeks (x)	1	2	3	4	5	6	7
Kilograms of litter collected (y)	2.9	2.5	2.5	2.3	2.1	1.9	1.7

a Plot the points from the table on a number plane.

b Graph the line of best fit whose equation is given by $y = -0.2x + 3$ on the same number plane.

c Use the equation of the line of best fit to find the number of kilograms of litter collected 12 weeks after the start of the anti-littering campaign.

10 The following table shows the number of eggs lain versus the number of ducks:

Ducks	1	2	3	4	5	6	7	8
Eggs	3	9	13	17	19	25	27	33

a Plot the points from the table on a number plane.

b Construct a line of best fit for the points on the same number plane.

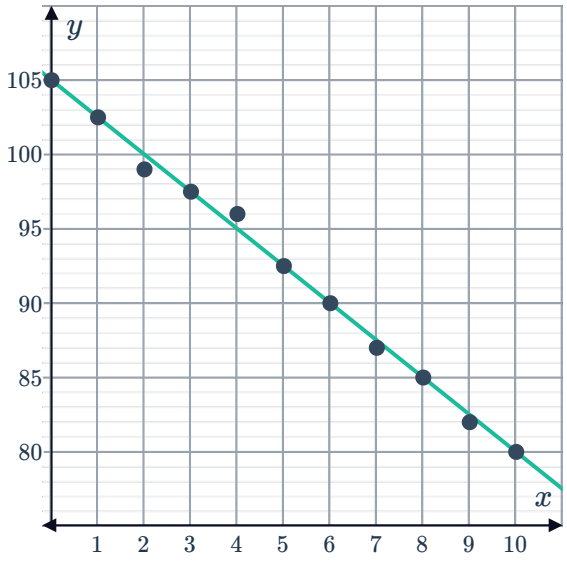
c Find the gradient of the line of best fit, if the line passes through $(3, 12)$ and $(1, 4)$.

d Find the y-intercept of this line of best fit.

e Find the equation of this line of best fit.

f Use this equation to find the number of eggs laid by 24 ducks.

11 The depth a diver, x, has descended below the surface of the water is plotted against her lung capacity, y:



a Does the line of best fit have a positive or negative gradient?

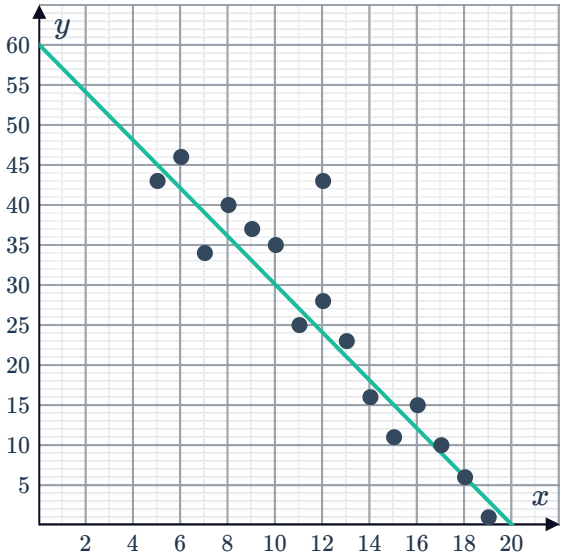
b Find the gradient of the line.

c Find the equation of the line of best fit.

d Use the line of best fit to estimate the lung capacity, y, at a depth of 4 metres.

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12 A number of people were asked how many hours each week, y, they spend on the internet. Their results were graphed against their age, x, in the scatter plot and a strong negative correlation was observed. A line of best fit has been drawn for the points:



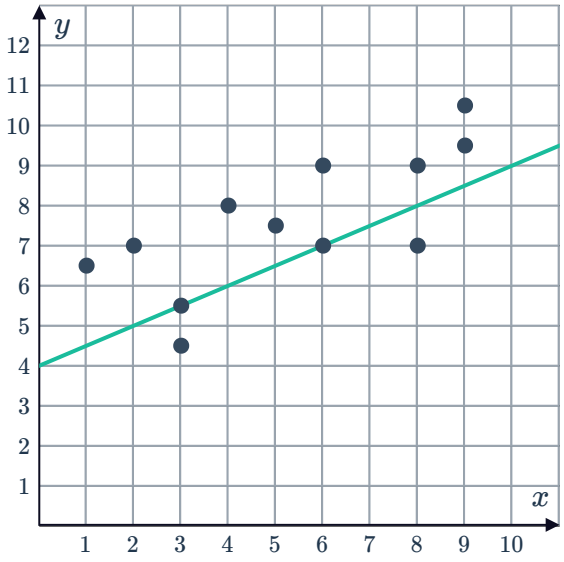
a Determine the x and y intercepts of the line of best fit.

b Using the intercepts, what is the gradient of the line of best fit?

c State the equation for the line of best fit in the form $y = mx + b$.

d Consider the response which is an outlier. According to the line of best fit, what would their usage be?

13 Consider the scatter plot shown:



a Using the two points on the line, determine the gradient of the line of best fit.

b Determine the equation of the line of best fit.

c Use the equation to approximate the value of y for $x = 6.9$.

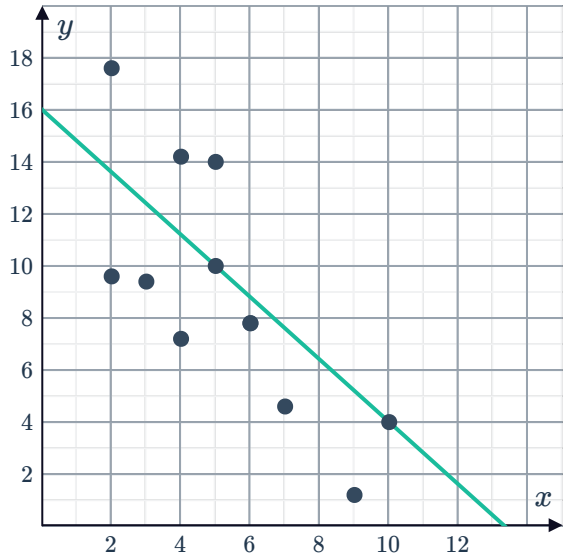
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14 Consider the scatter plot shown:

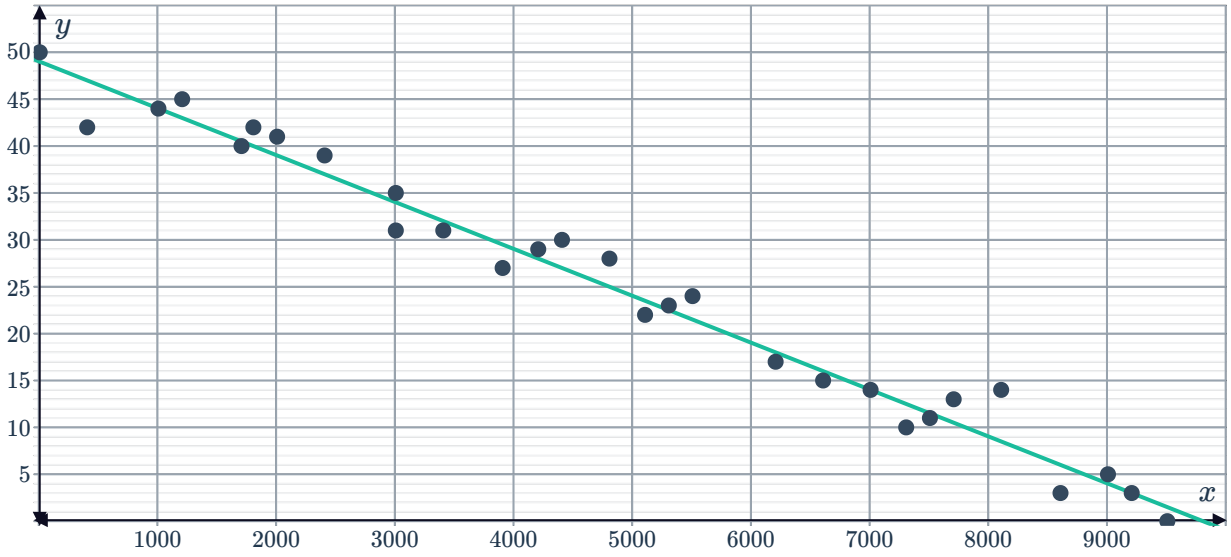
a Determine the gradient of the line of best fit.

b Find the y-intercept of the line of best fit.

c Determine the equation of the line of best fit.



15 The distance in kilometers, x, of several locations from the equator and their temperature in $^\circ C$, y, on a particular day is measured. The values are presented on the following scatter plot:

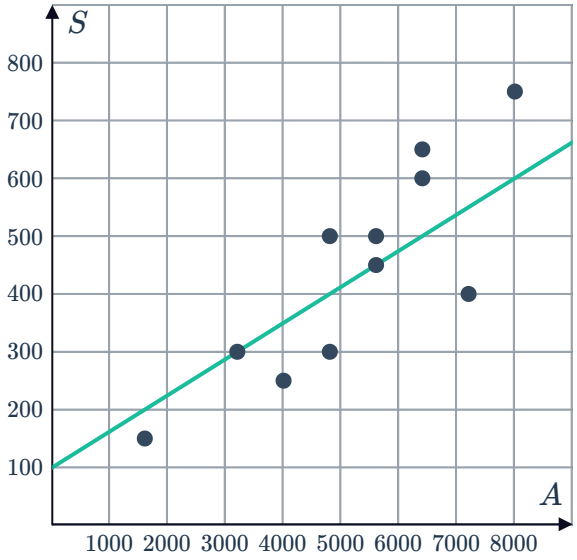


a Determine the equation of the line of best fit shown.

b Estimate the distance from the equator, x, if the temperature is $30.59^\circ \mathrm{C}$.

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16 A car company looked at the relationship between how much it had spent on advertising and the amount of sales each month over several months. The data has been plotted on the scatter graph and a line of best fit drawn:

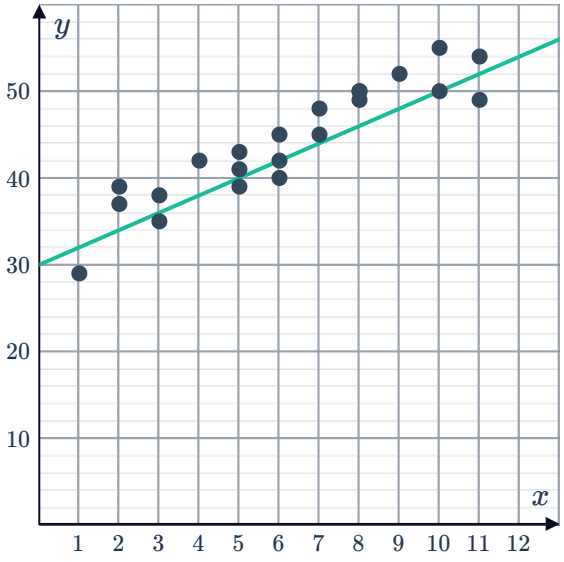


a Two points on the line are $(3200, 300)$ and $(5600, 450)$. Find the gradient of the line of best fit.

b The line of best fit can be written in the form $S = \frac{1}{16}A + b$, where S is the money spent on sales in thousands of dollars, and A is the advertising costs. Determine the value of b, the vertical intercept of the line.

c Use the line of best fit to estimate the number of sales next month if $\$4800$ is to be spent on advertising.

17 Several cars underwent a brake test and their age, x, was measured against their stopping distance, y. The scatter plot shows the results and a line of best fit that approximates the positive correlation:



a According to the line, what is the stopping distance of a car that is 6 years old?

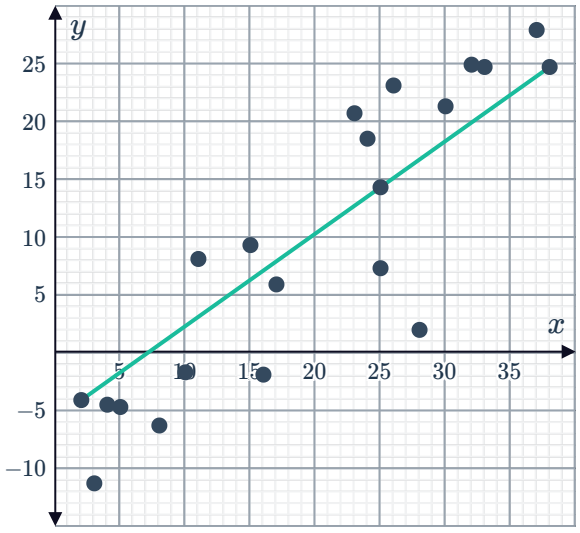
b Using the two marked points on the line, determine the gradient of the line of best fit.

c Determine the value of the vertical intercept of the line.

d Use the equation to estimate the stopping distance of a car that is 4.5 years old.

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18 The scatter plot shows the line of best fit for the relationship between air temperature, x, and sea temperature, y:
If the equation of the line of best fit is of the form $y = 0.8x + b$, determine the value of b.



19 The following table shows the temprature of a cooling metal versus the number of minutes that have passed:

Minutes (x)	1	2	3	4	5	6
Temperature (y)	27	27	23	23	19	19

a Plot the points from the table on a number plane.

b Graph the line of best fit on the same number plane.

c Find the gradient of the line of best fit, given that the line passes through $(5, 20)$ and $(3, 24)$.

d Find the y-intercept of the line of best fit.

e Find the equation of the line of best fit.

f Use the equation to find the number of minutes required to reach the temperature of 15°C.

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20 A study chose a few countries and measured the amount spent on healthcare per person each year, A, against the average life expectancy in that country, L:

A	L
100	27.5
400	44
1100	52.5
1500	71.5
2100	74.5

a Find one point on the line of best fit by taking the average of the values for each variable.

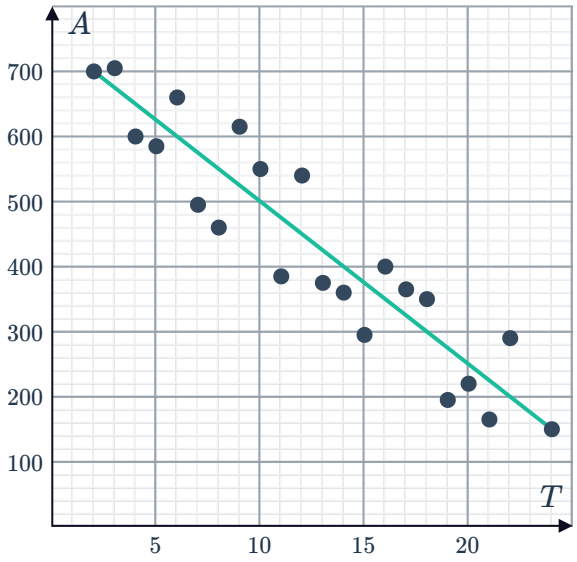
b Another point on the line of best fit is $(700, 45.5)$. Find the gradient of the line of best fit.

c Find the vertical intercept of the line.

d Find the equation of the line of best fit that relates A and L.

e A country is currently spending $\$30$ on healthcare per person each year. According to the line of best fit, by how much would the life expectancy of the country increase if healthcare spending is increased to $\$57$ per person each year?

21 A line of best fit has been drawn to approximate the relationship between sea temperature in $^\circ C$, T, and the area of healthy coral in hectares, A, in a particular location. Two particular points, $(2, 700)$ and $(24, 150)$, lie on the line.



a Find the gradient of the line.

b Find the vertical intercept of the line.

c Find the equation of the line of best fit.

d Using the line of best fit, find the sea temperature, T, at which there is predicted to be no healthy coral remaining.

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Determining the least squares line using technology

22 For each of the following sets of data:

- **i** Use technology to calculate the correlation coefficient to two decimal places.
- **ii** Describe the statistical relationship between the two variables.
- **iii** Use technology to form an equation for the least squares regression line. Round all values to one decimal place.

a

x	15.7	13.1	16.1	11	18.6	15.8	12.7	12.8	16.8	14.3
y	28.3	28.8	28.4	29	27.9	28.4	28.5	29	28.5	28.6

b

x	48.50	38.50	39.70	42.20	29.60	42.60	23.80	47.60	20.80
y	166.10	63.40	143.90	142.20	148.70	174.00	-52.30	195.50	65.50

23 The equation of the least squares regression line for a data set is given by $y = mx - 0.44$.

- **a** Given that the mean of x is 33 and the mean of y is 108.46, find the value of m.
- **b** Given that the $s_x = 114$ and $s_y = 396$, find the correlation coefficient, r.
- **c** Describe the strength of the relationship between x and y.

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